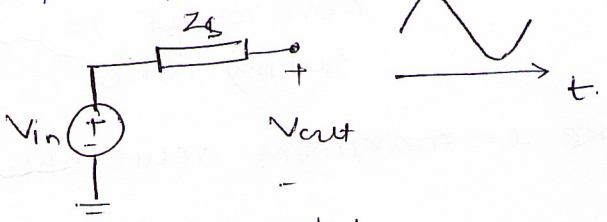
Differential Amplifier

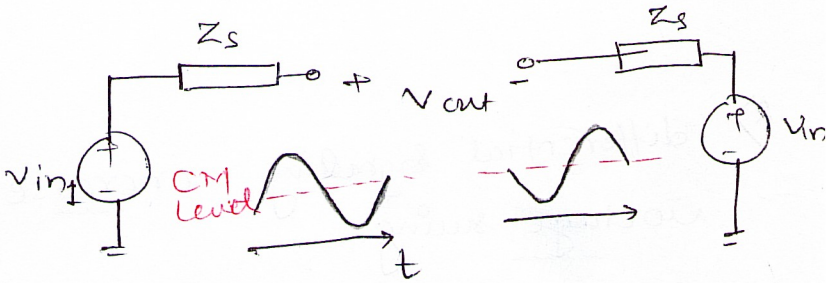
Single ended & Differential operation:-

→ A single ended signal is defined as one that is measured w.r.t a fixed potential, usually ground.

→ A differential signal is defined as one that is measured b/w two nodes that have equal & opposite signal excursion around a fixed potential.



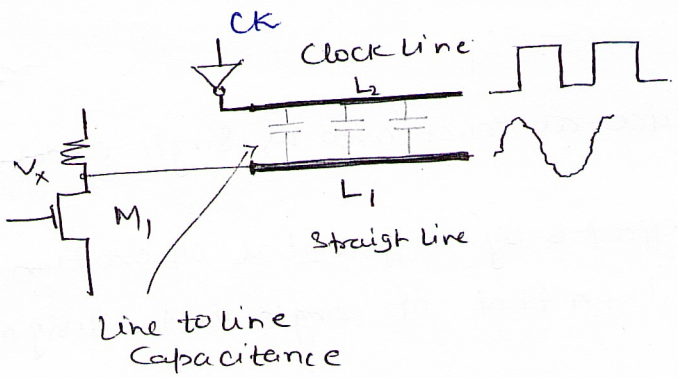
(a) single ended



→ the "center" potential in differential signalling is called common-mode level

Advantage of differential operation over single ended signalling

1) higher immunity to "environment" noise.

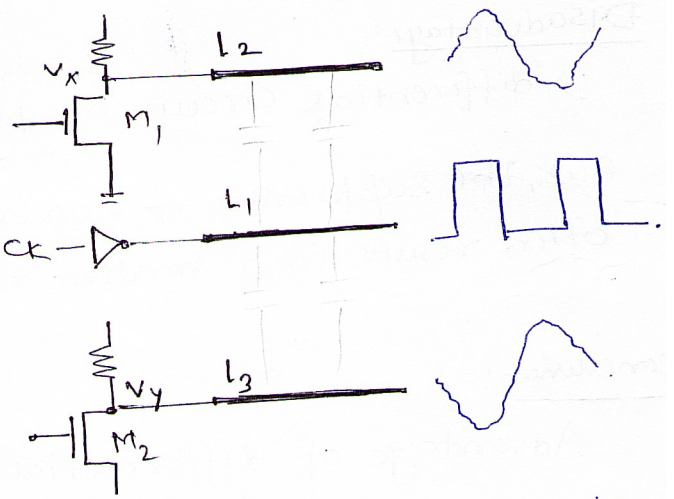


→ Due to capacitive coupling b/w lines, the transistion on line L_2 ~~Corrupt~~ the signal on line L_1.

Corrupt

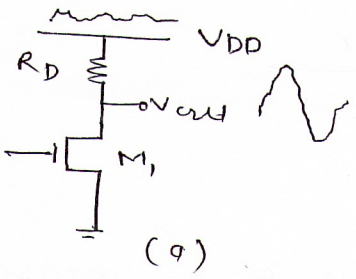
fig:- (a) corruption of signal due to coupling

Corruption

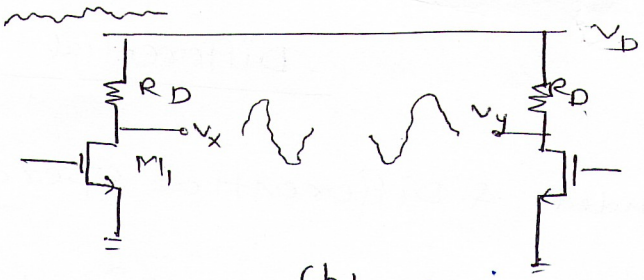


→ Clock line is placed midway b/w the two, the transition the differential phase by equal amount; leaving the difference since common mode of two phase is disturbed but the diff. o/p not co this arrangement reject comm mode noise

fig: (b) reduction of coupling by diff. op



(a)



(b)

Fig: Effect of supply noise on
(a) single ended circuit
(b) a differential circuit

$V_{out} = V_x - V_y$

↓

effect of noise will be lost

⇓

Circuit is much more robust to supply noise

2) differential signaling, increase ~~the~~ in maximum achievable voltage swing.

as in above fig. (b)

the max^m o/p swing at x ~~or~~ y ⇒ $V_{DD} - (V_{GS} - V_{TH})$

for $V_x - V_y$, the peak to peak swing is ⇒ $2[V_{DD} - (V_{GS} - V_{TH})]$

3) differential circuit. includes simpler biasing and higher linearity.

Disadvantage:-

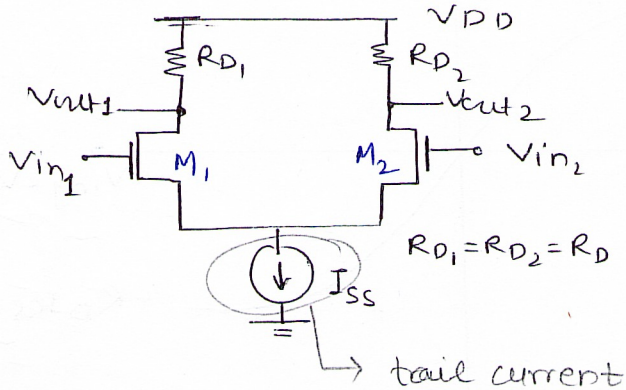
1) differential circuit occupy twice as much area as single-ended. But, the suppression of non ideal effects by differential operation often result in a smaller area than that of single ended design.

Conclusion:-

Advantage of differential signaling
1) high rejection of supply noise
2) high o/p swing

But what happens if V_{in1} and V_{in2} experience a large common-mode dc level? If the input cm level is excessively low, the minimum values of V_{in1} and V_{in2} may in fact turn off M_1 and M_2, leading to severe clipping at the o/p

Basic Differential pair:-



→ differential pair employs a current source I_{SS} to make $I_{D1} + I_{D2}$, independent of $V_{in,CM}$.

i.e. If $V_{in1} = V_{in2}$

→ bias current of each transister = $\frac{I_{SS}}{2}$

O/p common-mode level = $V_{DD} - R_D \frac{I_{SS}}{2}$

→ tail current, suppress the effect of I/p CM level variation on the operation of M_1 & M_2 and O/p level.

Qualitative Analysis:-

1) I/p - O/p characteristics of D.A:-

→ Assume $V_{in1} - V_{in2}$ → varies from $-\infty$ to ∞.

case-1: ⇒ If V_{in1} is more negative than V_{in2}, M_1 - Off, M_2 - On

∴ $I_{D2} = I_{SS}$

⇒ $V_{out1} = V_{DD}$, $V_{out2} = V_{DD} - I_{SS} R_D$

case-2:- V_{in1} is brought closer to V_{in2}, M_1 gradually turns on, drawing a fraction of I_{SS} from R_{D1} and hence lowering V_{out1}.

case-3:- $V_{in1} = V_{in2}$, ⇒ $V_{out1} = V_{out2} = V_{DD} - R I_{SS}/2$

case4:- Now, V_{in1} become more positive than V_{in2} M_1 carries greater current than does M_2 and V_{out1} drop below V_{out2} And for sufficiently large $V_{in1} - V_{in2}$, M_1 hogs all of I_{SS} turning M_2 off. As a result,

"Hogs"

$V_{out1} = V_{DD} - R_D I_{SS}$

$V_{out2} = V_{DD}$

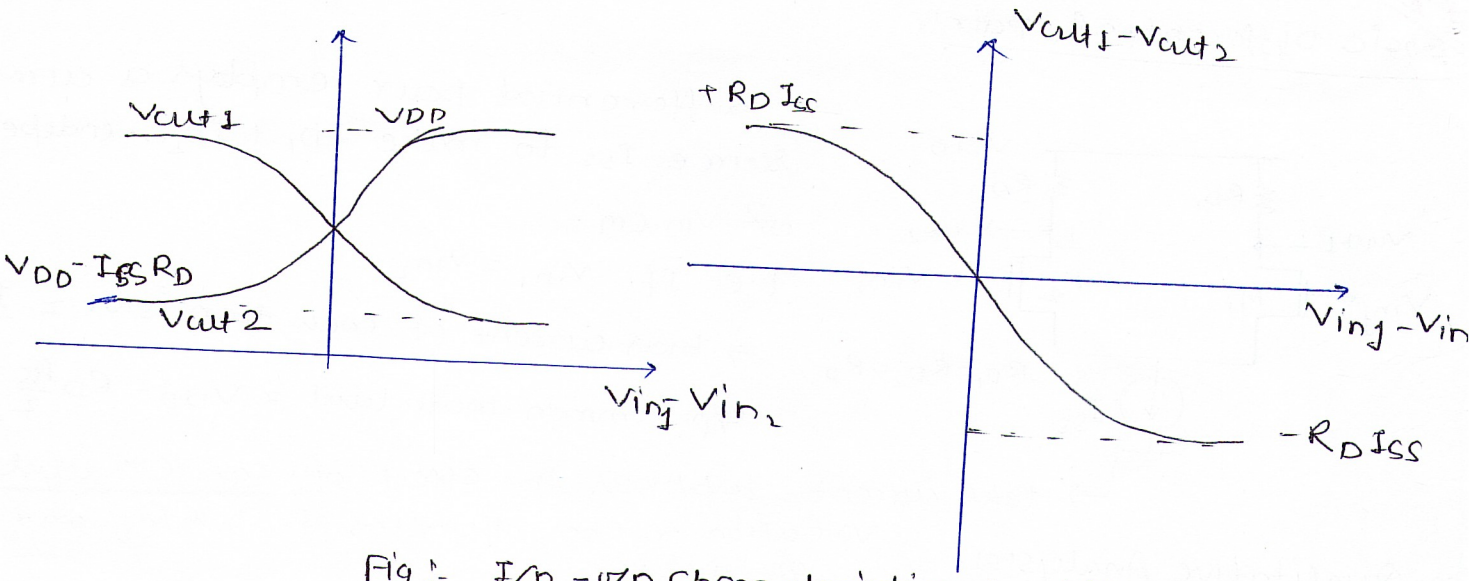


Fig :- I/p - O/p characteristics of differential pair.

Above analysis attributes of ~~two~~ important outcome:-

i) maximum and minimum levels at the O/p are well defined i.e. V_{DD} & $V_{DD} - R_D I_{SS}$.

ii) Independent of I/p CM level.

iii) the small signal gain (slope of $V_{out1} - V_{out2}$ Vs $V_{in1} - V_{in2}$) is maximum for $V_{in1} = V_{in2}$, gradually falling to zero as $|V_{in1} - V_{in2}|$ increases. ✱

for $V_{in1} = V_{in2}$, we can say circuit is at equilibrium.

Q:- What should be the range of CM I/p voltage to operate device for the effect of CM level variation on the operation of M_1 and M_2 and O/p level at lowest level or negligible?

Consider basic differential pair circuit with I_{SS} implemented by an NFET.

→ We set $V_{in1} = V_{in2} = V_{in,CM}$

& $V_{in,CM}$ varies from 0 to V_{DD}

(✱ In other words, the circuit becomes more nonlinear as the input voltage swing increases.)

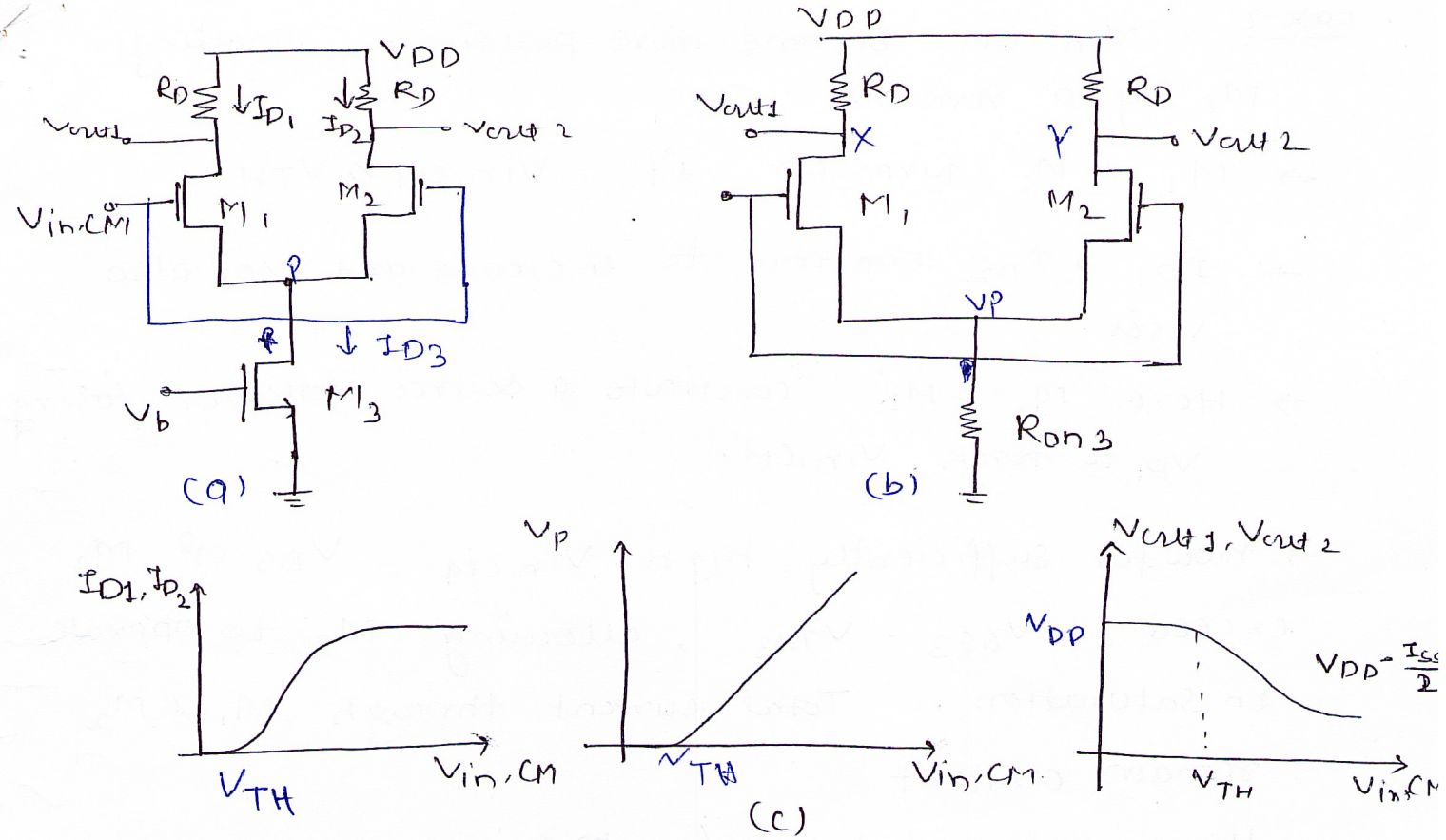


Fig:- (a) Differential pair sensing i/p CM change

(b) equivalent circuit if M_3 operates in deep triode region

(c) CM i/p - o/p characteristics

Analysis:- from above circuit (a) & (b)

Case-1

if $V_{in,CM} = 0$ $\Rightarrow$ M_1, M_2 $\rightarrow$ off

$\Rightarrow$ $I_{D3} = 0$

$\Rightarrow$ M_3 is in deep triode region.

because V_b is high enough to create an inversion layer in the transistor.

$\Rightarrow$ $I_{D1} = I_{D2} = 0$, no amplification

$\Rightarrow$ $V_{out1} = V_{out2} = V_{DD}$

case-2 $V_{in,CM}$ become more positive. Modeling M_2 by a resister.

→ M_1 & M_2 turns on if $V_{in,CM} \geq V_{TH}$

→ I_{D1} & I_{D2} continue to increase and V_P also rises.

→ Here, M_1 & M_2 constitute a source follower, forcing V_P to track $V_{in,CM}$.

→ now for sufficiently high $V_{in,CM}$, V_{DS} of M_3 exceed, $V_{GS3} - V_{TH3}$, allowing M_3 to operate in saturation. Total current through M_1 & M_2 remain constant

Hence we conclude that for proper operation,

$$V_{in,CM} \geq V_{GS1} + (V_{GS3} - V_{TH3})$$

case-3 If $V_{in,CM}$ further rises, since V_{out1} & V_{out2} relatively constant, we expect that M_1 & M_2 enter the triode region if,

$$V_{in,CM} > V_{out1} + V_{TH} = V_{DD} - R_D \frac{I_{SS}}{2} + V_{TH}$$

Hence

allowable value of $V_{in,CM}$ is bounded as follow,

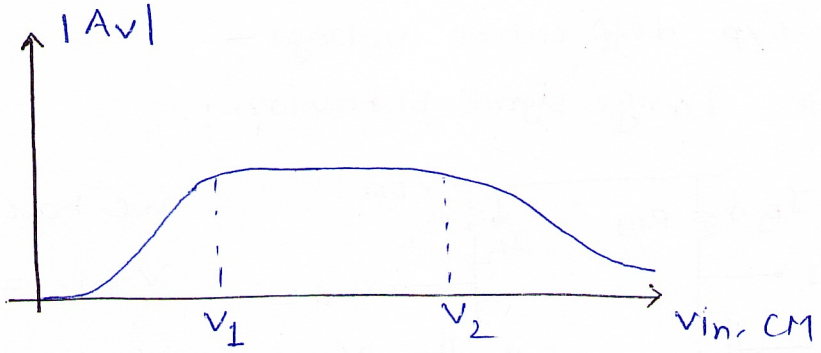
**

$$V_{GS1} + (\cancel{V_{GS3}} V_{GS3} - V_{TH3}) \leq V_{in,CM} \leq \min\left[V_{DD} - R_D \frac{I_{SS}}{2} + V_{TH}, V_{DD}\right]$$

*** Imp

Small signal gain (differential gain) (A_v) as a function of i/p CM level :-

Graph:-



Explanation:-

→ The gain begins to increase as $V_{in,CM}$ exceeds V_{TH}. After tail current source (I_{SS}), enters saturation. ($V_{in,CM} = V_1$), the gain relatively constant, finally if $V_{in,CM}$ is very high the input transistor enter the triode region ($V_{in,CM} = V_2$) gain begin to fall.

Conclusion:-

→ For M_1, & M_2 to be saturated, each o/p can go as high as V_{DD} but as low as approximately $V_{in,CM} - V_T$

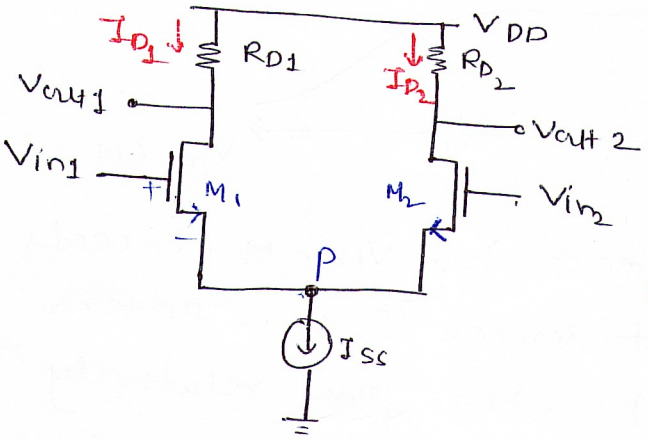
i.e. higher the i/p CM level, the smaller the allowable o/p swing. For this reason, it is desirable to choose a relatively low $V_{in,CM}$.

→ The gain of differential pair is a function of the DC drop across the load resistor. Thus, if $\frac{R_D I_{SS}}{2}$ is large, $V_{in,CM}$ must remain close to ground potential.

Quantitative Analysis:-

→ Behavior of a MOS differential pair as a function of the i/p differential voltage –

Consider → Large signal behavior:-



we have,

$$V_{out1} = V_{DD} - R_{D1} I_{D1}$$

$$V_{out2} = V_{DD} - R_{D_2} I_{D_2}$$

i.e.

$$V_{out1} - V_{out2} = R_{D_2} I_{D_2} - R_{D_1} I_{D_1}$$

$$= R_D (I_{D_2} - I_{D_1})$$

Calculation of I_{D_1} & I_{D_2} in terms of V_{in1} & V_{in_2} if $R_{D_1} = R_{D_2} = R_D$

→ assume, symmetric ckt, M_1 & M_2 are saturated & $\lambda = 0$

Voltage @ node 'P' V_P $V_{in1} - V_{GS1}$ & $V_{in2} - V_{GS_2}$

$$\therefore V_{in1} - V_{in_2} = V_{GS1} - V_{GS_2} \quad \text{—— (1)}$$

we know,

$$I_{D_{sat}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

$$\therefore (V_{GS} - V_{TH})^2 = \frac{I_D}{\frac{1}{2} \mu_n C_{ox} W/L}$$

$$\therefore V_{GS} = \sqrt{\frac{2 I_D}{\frac{1}{2} \mu_n C_{ox} W/L}} + V_{TH} \quad \text{—— (2)}$$

Hence from (1) & (2)

$$V_{in1} - V_{in_2} = \sqrt{\frac{I_{D1}}{\frac{1}{2} \mu_n C_{ox} \frac{W}{L}}} - \sqrt{\frac{I_{D_2}}{\frac{1}{2} \mu_n C_{ox} \frac{W}{L}}}$$

$$\Rightarrow V_{in_1} - V_{in_2} = \sqrt{\frac{2 I_{D1}}{\mu_n C_{ox} \frac{W}{L}}} - \sqrt{\frac{2 I_{D2}}{\mu_n C_{ox} \frac{W}{L}}} \quad \text{— (3)}$$

~~Squadre~~ Square eq (3) and put $I_{D_1} + I_{D_2} = I_{SS}$ we will get

$$(V_{in_1} - V_{in_2})^2 = \frac{2}{\mu_n C_{ox} \frac{W}{L}} \left(I_{SS} - 2\sqrt{I_{D_1} \cdot I_{D_2}} \right)$$

$$\Rightarrow \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{in_2})^2 = I_{SS} = -2\sqrt{I_{D_1} \cdot I_{D_2}} \quad \text{— (4)}$$

Squaring eq (4) and ~~base~~,

$$4 I_{D_1} I_{D_2} = (I_{D_1} + I_{D_2})^2 - (I_{D_1} \ominus I_{D_2})^2$$

$$= I_{SS}^2 - (I_{D_1} - I_{D_2})^2$$

from eq (4)

$$\therefore (I_{D1} - I_{D2})^2 = -\frac{1}{4} \left(\mu_n C_{ox} \frac{W}{L}\right)^2 (V_{in_1} - V_{in_2})^4 + I_{SS} \mu_n C_{ox} \frac{W}{L} (V_{in_1} - V_{in_2})^2$$

$$\therefore \boxed{(I_{D1} - I_{D2}) = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in_1} - V_{in_2}) \cdot \sqrt{\frac{4 I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - (V_{in_1} - V_{in_2})^2}}$$

i.e. if $V_{in_1} = V_{in_2} \Rightarrow I_{D_1} - I_{D_2} = 0$

$\Rightarrow I_{D_1} - I_{D_2}$ is an odd function of $V_{in_1} - V_{in_2}$.

calculation of transconductance (g_m) & voltage gain (A_V) :-

equivalent G_m of M_1 & M_2

$$G_m = \frac{\partial \Delta I_D}{\partial \Delta V_{in}}$$

take derivative of eq (4)

**

$$G_m = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \frac{\frac{4 I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - 2\Delta V_{in}^2}{\sqrt{\frac{4 I_{SS}}{\mu_n C_{ox} \frac{W}{L}} - \Delta V_{in}^2}} \quad \text{— (5)}$$

for $\Delta V_{in} = 0$

**

$$G_m = \sqrt{\mu_n C_{ox} \left(\frac{W}{L}\right) I_{SS}} \quad \text{— (6)}$$

$$V_{out1} - V_{out2} = R_D \cdot \Delta I = R_D \cdot G_m \Delta V_{in}$$

∴ small signal differential voltage gain of the circuit, in the equilibrium condition,

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$$|A_V| = \sqrt{\mu_n C_{ox} \left(\frac{W}{L}\right) I_{SS}} \cdot R_D \quad \text{— (7)}$$

eq (5) can also be zero if

**

$$\frac{\Delta V_{in}}{\Delta V_{in}} = \sqrt{\frac{2 I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} \quad \text{— (8)}$$

maximum differential i/p that circuit can handle.

In reality as ΔV_{in} exceeds a limit, one transistor carries the entire I_{SS}, turning off the other. Denoting this value by ΔV_{in1}, we have $I_{D1} = I_{SS}$ and $\Delta V_{in} = V_{in}$

because M_2 is nearly off.

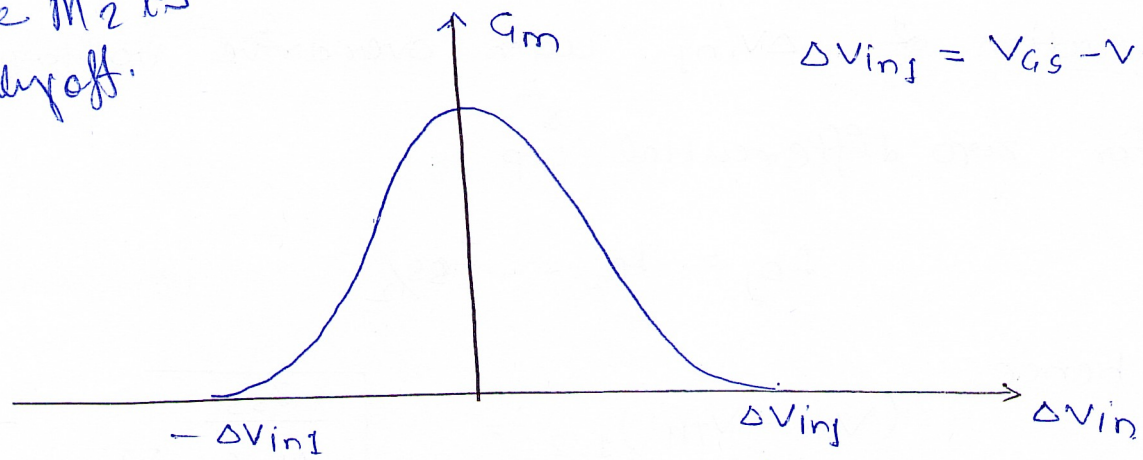


Fig:- transconducantence vs i/p voltage.

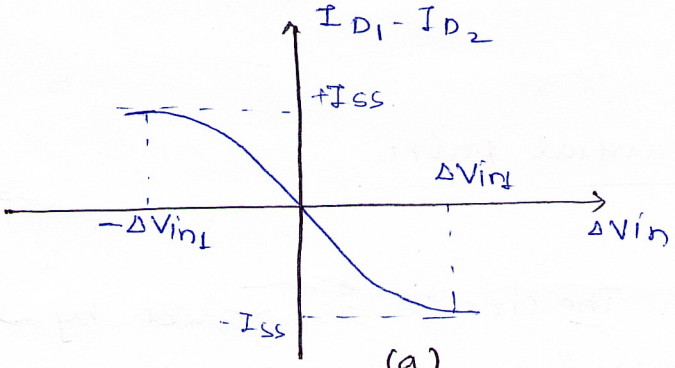
$G_m = 0$ for $\Delta V_{in} = \Delta V_{in1}$.

Exp:- Plot i/p - o/p characteristics of a differential pair as device width and tail curren (I_{SS}) vary.

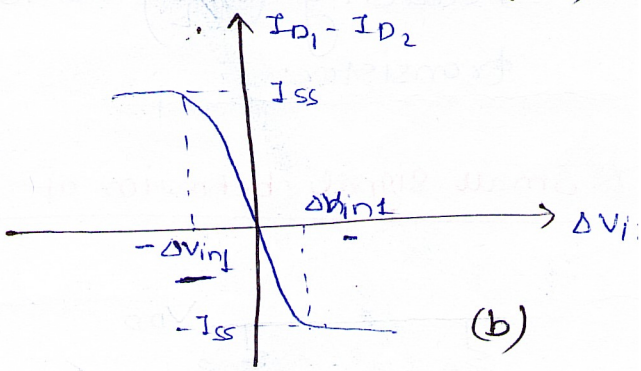
$$\because \quad \Delta V_{in1} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox} \frac{W}{L}}}$$

Case:-1

as $\Delta V_{in} \downarrow \Rightarrow \left(\frac{W}{L}\right) \uparrow \Rightarrow$ narrowing the i/p range fig.(b)



(a)

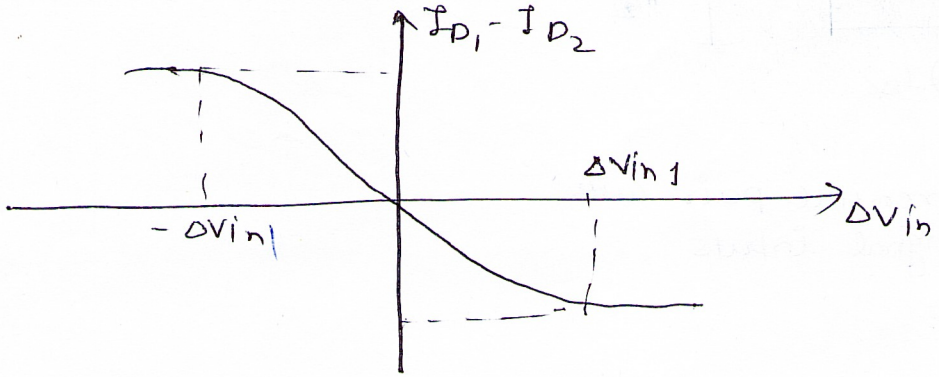


(b)

(narrow i/p range)

Case-2

as $I_{SS} \uparrow \Rightarrow$ both i/p range and o/p current swing increase fig.(c)



Relation of ΔV_{in_1} with overdrive voltage:-

For zero differential i/p, M_1 & M_2 @ equilibrium

$$I_{D_1} = I_{D_2} = I_{SS}/2$$

hence

$$(V_{GS} - V_{TH})_{1,2} = \sqrt{\frac{I_{SS}}{\mu_n C_{ox} \frac{W}{L}}} \quad \text{---} \quad (9)$$

from eq (8) & (9)

$$\text{equilibrium overdrive voltage} = \frac{\Delta V_{in_1}}{\sqrt{2}}$$

Conclusion:-

→ Increasing ΔV_{in_1} make circuit more linear inevitably increase overdrive voltage of M_1 & M_2 and given I_{SS}. This can accomplish only by reducing (W/L) and hence transconductance of transister.

Small signal behavior of differential pair:-

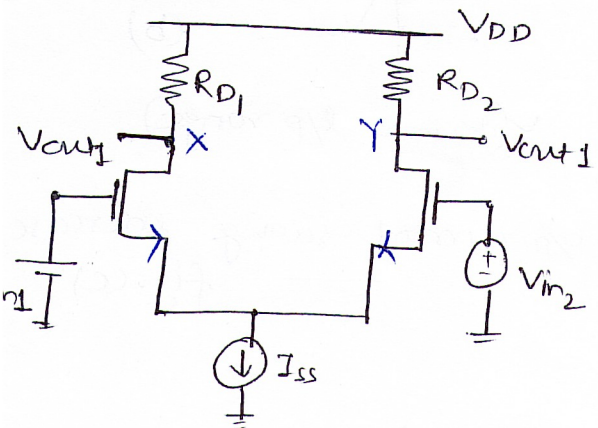
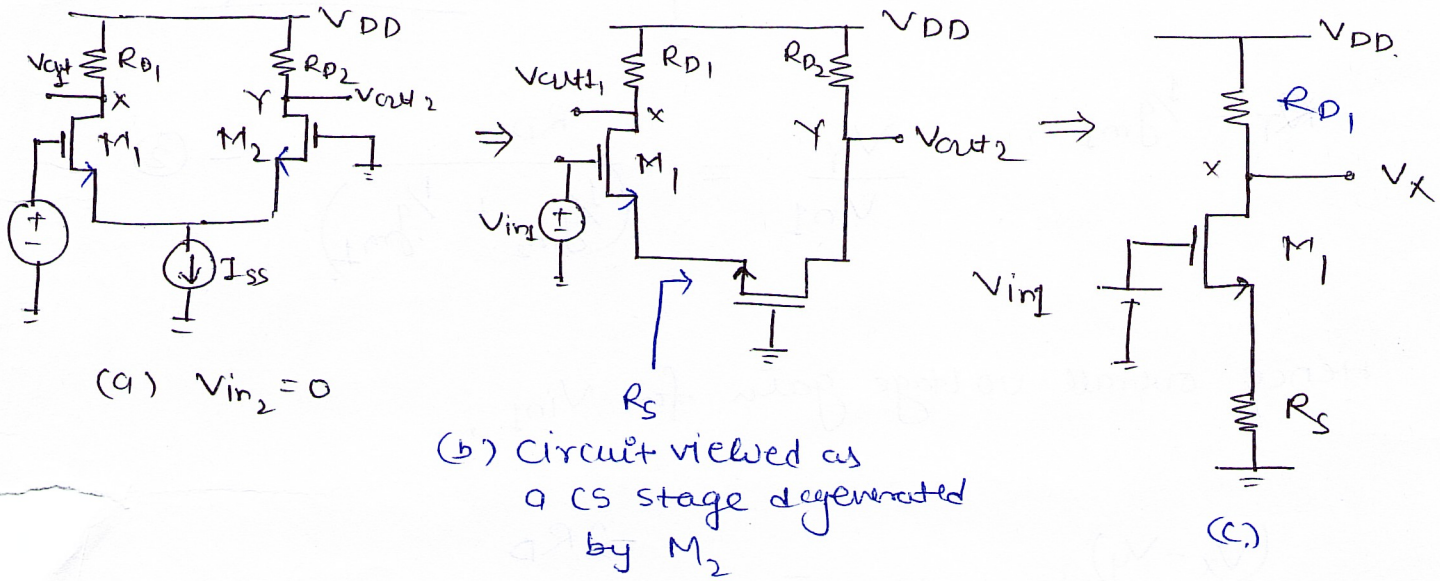


Fig:- Differential pair with small signal inputs

→ The circuit is driven by two independent signals. Thus o/p can be computed by super-position.

Case-1:- $V_{in_2} = 0$ and find effect of V_{in1} at X & Y

To obtain V_X, M_1 forms a common source stage with degeneration resistance R_S equal to impedence seen looking into the source of M_2.



(a) $V_{in_2} = 0$

(b) Circuit viewed as a CS stage degenerated by M_2

(c)

Neglect the channel length Modulation & body effect,

we have $R_S = \frac{1}{g_{m_2}}$ $\qquad R_{D_1} = R_{D_2} = R_D$

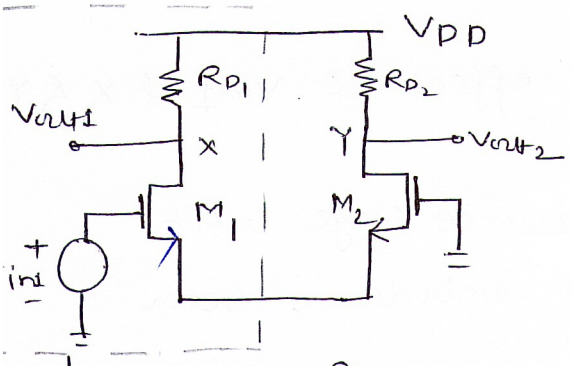
Hence,

$$\frac{V_x}{V_{in_1}} = \frac{-R_D}{\left(\frac{1}{g_{m_1}} + \frac{1}{g_{m_2}}\right)} \qquad \text{—①}$$

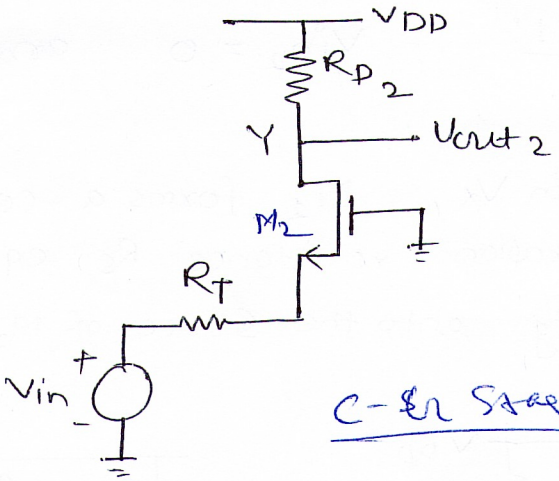
now to calculate V_y, M_1 drive M_2 as a source follower and replace V_{in1} and M_1 by thevenin's equivalent

and $V_{Thevenin} = V_T = V_{in}$ $\qquad R_{th} = R_T = \frac{1}{g_{m_1}}$

Hence, M_2 operates as a common-gate stage, exhibiting a gain equal to →



↳ Replace with Thevinin equivalut



C-S Stage

$R_T = 1/g_{m1}$, $\frac{V_Y}{V_{in1}} = \frac{R_D}{(1/g_{m2} + 1/g_{m1})}$ — (2)

Hence overall voltage gain for V_{in1},

$$(V_x - V_y)_{\text{due to } V_{in1}} = \frac{-2R_D}{(1/g_{m1} + 1/g_{m2})} V_{in1} \quad — (3)$$

if $g_{m1} = g_{m2} = g_m$

$$(V_x - V_y)_{\text{Dut to } V_{in1}} = -g_m R_D V_{in1} \quad — (4)$$

Similarly by virtue of symmetry, the effect of V_{in2} at X & Y is identical to that of V_{in1} except for a ~~change~~ change in polarities:

$$(V_x - V_y)_{\text{Dut to } V_{in2}} = -g_m R_D V_{in2} \quad — (5)$$

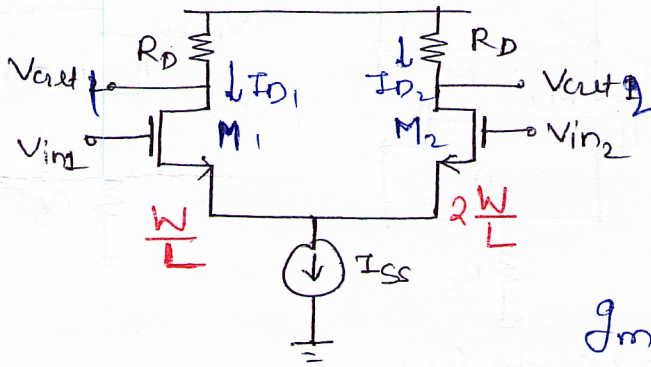
from eq (4) & (5)

$$A_V = \frac{(V_x - V_y)_{total}}{V_{in1} - V_{in2}} = -g_m R_D$$

$$|A_{v1}| = g_m R_D$$

Example-2

In following circuit, M_2 is twice as ~~wide~~ wide as M_1. Calculate the small signal gain if the bias values of V_{in1} and V_{in2} are equal.



$\because$ M_1 and M_2 are at same potential, $V_{GS1} = V_{GS2}$

$$I_{D2} = 2\, I_{D1} = 2\,\frac{I_{SS}}{3}$$

$$g_{m_1} = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)\frac{I_{SS}}{3}}$$

$$g_{m_2} = \sqrt{2\mu_n C_{ox}\left(\frac{2W}{L}\right)\frac{2I_{DS}}{3}} = 2g_{m_1}$$

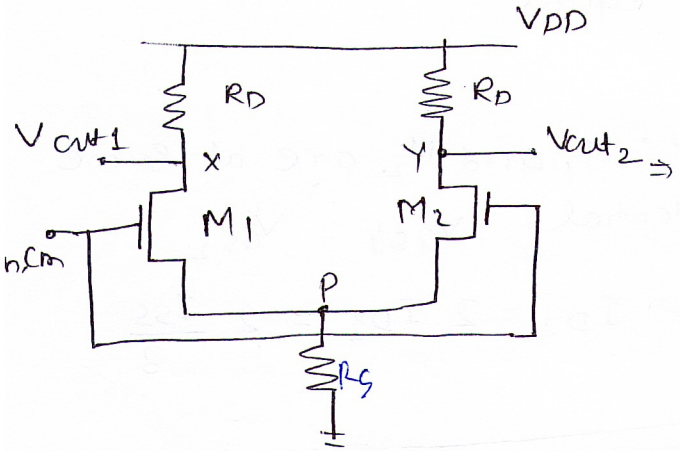
from above derivation, we can write,

$$|A_{v1}| = \frac{2R_D}{1/g_{m_1} + \frac{1}{2g_{m_1}}} = \frac{4}{3} g_{m_1} R_D$$

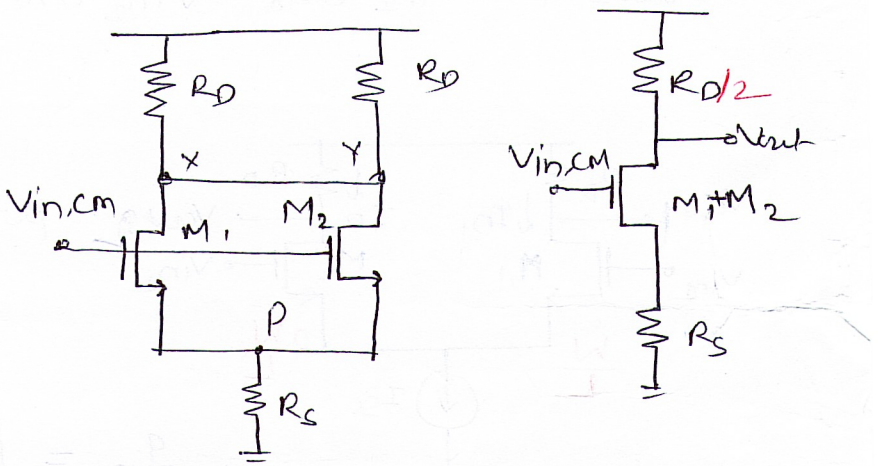
Common-Mode Response:- (of Differential Pair)

→ In case of differential amplifier, it has ability to suppress the effect of common mode perturbation.

→ But, In reality, neither is the circuit fully symmetric nor does the current source exhibit infinit output impedance As a result, a fraction of the input CM variation appears at the o/p.



(a) Differential pair sensing common mode i/p



(b) Simplified ckt

→ considering symmetry, ~~Vx remain~~

→ As Vin,CM changes, so does Vp, there by increasing drain current of M_1 and M_2 and lowering Vx and Vy. Due to symmetry Vx and Vy equals, hece two node can be shorted together as in above fig. (b)

→ compound device, $M_1 + M_2$ has twice the width and the bias current of each M_1 & M_2 and therefore twice their transconductence.

The common mode gain of circuit,

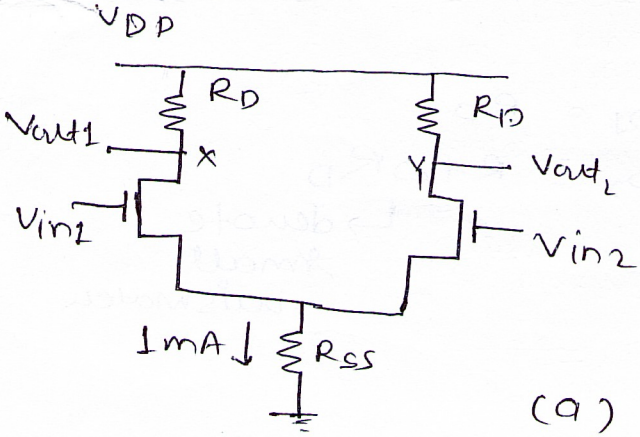
$$A_{v\,CM} = \frac{V_{out}}{V_{in.\,CM}}$$

$$\boxed{A_{v,CM} = -\frac{(R_D/2)}{\left(\frac{1}{2g_m} + R_{SS}\right)}} \quad \text{—(1)}$$

$g_m \rightarrow$ transconductance of each M_1 and M_2 and $\lambda = \gamma = 0$

Unsolved Example:-

Following circuit uses resister rather than a current source to define tail current of 1 mA.



Assume,

$(W/L)_{1,2} = 25/0.5$ $\mu_n C_{ox} = 50\,\mu A/V^2$

$V_{TH} = 0.6V$ $\lambda = \gamma = 0$

$V_{DD} = 3V$

(a) what is required i/p CM for which R_{SS} sustain 0.5 V?

(b) Calculate R_D for a differential gain of 5

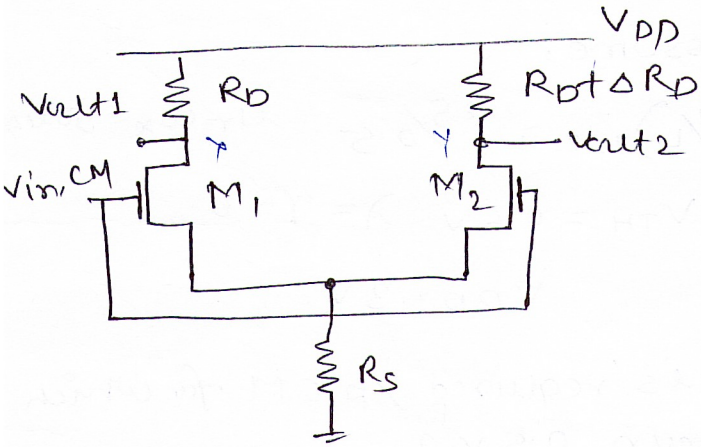
(c) What happens at the o/p if the i/p CM level is 50mV higher than the value calculate in (a).

→ from above common mode gain, which indicate that finite o/p impedance of tail current source result in some common-mode gain in a symmetrical differential amplifier. ~~pair~~.

→ The problem is that, the variation of the differential o/p as the result of a change in $V_{in,CM}$, an affect of occur because in reality the circuit is not fully symmetric. i.e. the two sides slightly mis-matches during manufacturing.

→ Effect of i/p common mode variation if the circuit is asymmetric:-

→ tail current source suffers from finit output impedance.



$R_{D1} = R_D$

$R_{D2} = R + \Delta R_D$

↳ denote small mismatch

(a) common mode response in presence of resistor mismatch

→ Since M_1 and M_2 are identical, I_{D_1} & I_{D_2} increase by $\left(\frac{g_m}{1+2g_m R_{SS}}\right)\Delta V_{in,CM}$, but ~~Vx~~ V_x and V_y change by different amount.

$$\Delta V_x = -\Delta V_{in,CM}\left(\frac{g_m}{1+2g_m R_{SS}}\right) R_D.$$

$$\Delta V_y = -\Delta V_{in,CM}\left(\frac{g_m}{1+2g_m R_{SS}}\right)(R_D + \Delta R_D)$$

↓ differential component

→ A common mode change at the i/p introduce a differentia component at the o/p. The circuit exhibit common-mo to differential conversion.

→ i/p of differential pair include both a differential signal and common mode noise, the circuit corrupt the amplified differential signal by the i/p CM change.

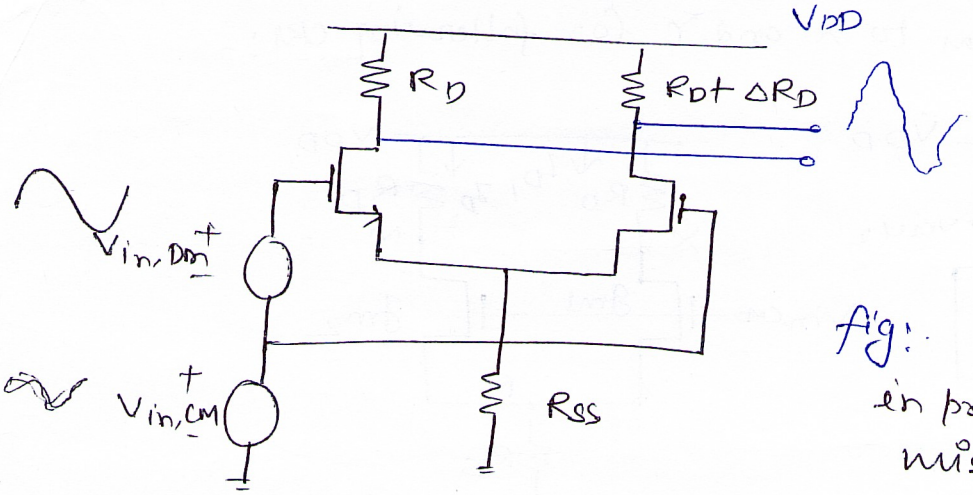


fig: Effect of CM noise in presence of resistor mismatch.

→ Hence we can say, the common-mode response of differential pairs depends on the o/p impedance of the tail current source and asymmetric in circuit, it causes two effects (i) variation of the o/p CM level (in absence of mismatch) and (ii) conversion of i/p common-mode variation to differential component at the o/p.

~~Ass~~

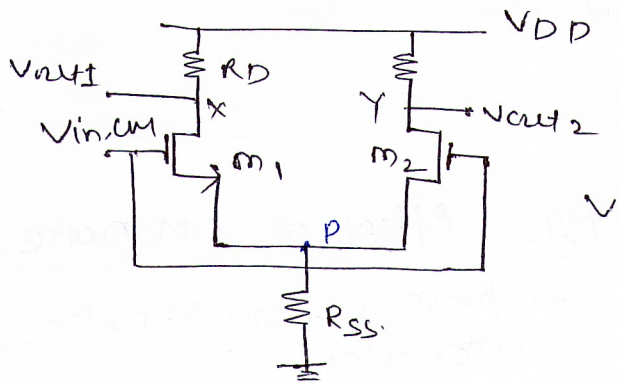
Note:- Even if the o/p resistance of the current source is high, common-mode to differential conversion become significant at high frequency. ~~This~~ There is effect of capacitance arises from the parasitic of the current source itself as well as the source but junction of M_1 and M_2.

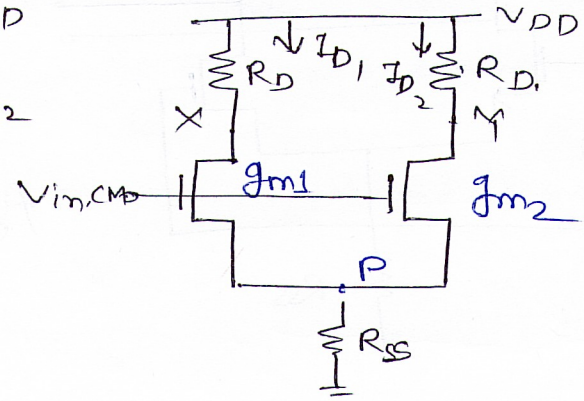
Assymetric from mismatch between M_1 & M_2 :-

→ It owing to dimension and threshold voltage mismatch.

→ The two transister exhibit different current and unequal transconductance.

Calculation of $V_{in,CM}$ to X and Y for following ckt:-





$$I_{D_1} = g_{m_1}(V_{in,CM} - V_P) \qquad I_{D_2} = g_{m_2}(V_{in,CM} - V_P)$$

i.e.

$$(g_{m_1} + g_{m_2})(V_{in,CM} - V_P) R_{SS} = V_P$$

$$V_p = \frac{(g_{m_1} + g_{m_2}) R_{SS}}{(g_{m_1} + g_{m_2}) R_S + 1} V_{in;cm}$$

now, o/p voltage,

$$V_X = -g_{m_1} (V_{in,CM} - V_p) R_D$$

from above value of V_p,

$$V_X = -g_{m_1} / (g_{m_1} + g_{m_2}) R_{SS} + 1 \times R_D V_{in,CM}$$

$$V_y = -g_{m_2} (V_{in,CM} - V_p) R_D$$

$$= \frac{-g_{m_2}}{(g_{m_1} + g_{m_2}) R_{SS} + 1} R_D \cdot V_{in,CM}$$

differential o/p

$$V_X - V_y = -\left(\frac{g_{m_1} - g_{m_2}}{(g_{m_1} + g_{m_2}) R_{SS} + 1}\right) R_D \cdot V_{in,CM}$$

$$\frac{V_X - V_y}{V_{in,CM}} = A_{CM-DM} = -\frac{\Delta g_m R_D}{(g_{m_1} + g_{m_2}) R_{SS} + 1}$$

where A_{CM-DM} → CM to DM conversion

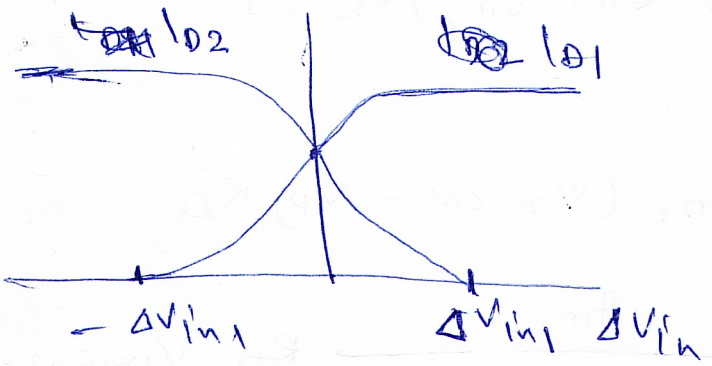
$$\Delta g_m = g_{m_1} - g_{m_2}$$

Common mode rejection ratio (CMRR)

defined as,

$$CMRR = \left| \frac{A_{DM}}{A_{CM-DM}} \right|$$

$$A_{DM} = g_{m(equivalent)} \cdot R_D$$



If only g_m mismatch is considered, then

$$|A_{DM}| = \frac{R_D}{2} \frac{g_{m1} + g_{m2} + 4 g_{m1} g_{m2} R_{SS}}{1 + (g_{m1} + g_{m2}) R_{SS}},$$

where it is assumed $V_{in1} = -V_{in2}$, and hence

$$CMRR = \frac{g_{m1} + g_{m2} + 4 g_{m1} g_{m2} R_{SS}}{2 \Delta g_m}$$

$$\approx \frac{g_m}{\Delta g_m} (1 + 2 g_m R_{SS}).$$

Where g_m denotes the mean value, i.e $g_m = \frac{g_{m1} + g_{m2}}{2}$

In practice, all mismatches must be taken into account.